Lecture 4

Order and completeness in sequence limits

Uses.

Summary 2: A supremum is a least upper bound. Every nonempty real set bounded above has one (R). Bernoulli gives $(1+h)^n \geq 1+nh$ for $h \geq -1$. The Archimedean property supplies integer cutoffs. Standing fact E supplies the school trigonometric bounds and positive-base powers.
Summary 3: Convergence means an error below each positive tolerance eventually. Limits are unique; sums, products and quotients with nonzero denominator limit obey the limit laws. For real $|q| < 1$, $q^n \to 0$.

Objectives. *An error bound can prove a proposed limit. Order and completeness also prove that a limit exists before its value is known. We use a supremum to prove monotone convergence, construct the number e, and prove that every bounded real sequence has a convergent subsequence.*

By the end of this lecture you will be able to:

- Pass an eventual inequality to limits and apply the squeeze theorem.
- Prove convergence of a monotone sequence using a bound.
- Construct e between an increasing and a decreasing sequence.
- Extract a convergent subsequence from a bounded real sequence.

4.1 Order and the squeeze theorem

We say that $a_n \leq b_n$ *eventually* if the inequality holds for all $n \geq N_0$ for some $N_0 \in \mathbb{N}$. The first terms need not satisfy it. Only the later terms enter a convergence proof.

Proposition 4.1 (Order passes to limits). *Suppose $a_n \to a$, $b_n \to b$, and $a_n \leq b_n$ eventually. Then $a \leq b$.*

Proof. Suppose instead that $a > b$. Put $\varepsilon = (a-b)/3 > 0$. For every sufficiently large n, convergence supplies $|a_n - a| < \varepsilon$ and $|b_n - b| < \varepsilon$. The eventual order also holds at such an index. But

$$b_n < b + \varepsilon < a - \varepsilon < a_n.$$

The middle inequality follows from $a - b = 3\varepsilon$. This contradicts $a_n \leq b_n$. Hence $a \leq b$. ∎

Strict inequalities need not stay strict in the limit. For example, $0 < 1/n$ for every $n \in \mathbb{N}$, but both sides converge to zero. The proposition concludes a non-strict inequality, even when every term comparison is strict.

Theorem 4.2 (Squeeze theorem). *Suppose $a_n \le b_n \le c_n$ eventually and both $a_n \to L$ and $c_n \to L$. Then $b_n \to L$.*

Proof. Let $\varepsilon > 0$ be given. Choose one cutoff after which both outside errors are below ε and both inequalities hold. For every n at or after that cutoff,

$$L - \varepsilon < a_n \le b_n \le c_n < L + \varepsilon.$$

Thus $|b_n - L| < \varepsilon$. This is the definition of $b_n \to L$. ∎

The theorem does not assume that the middle sequence has a limit. It proves its convergence from the two outside estimates. Their limits must agree for this conclusion.

Example 4.3 (An oscillating numerator). Consider the sequence $b_n = (\sin n)/n$, with the angle in radians. Standing fact E gives $-1 \le \sin n \le 1$. Since $n > 0$,

$$-\frac{1}{n} \le b_n \le \frac{1}{n}.$$

Both outside sequences tend to zero by Lecture 3. The squeeze theorem therefore gives $b_n \to 0$. No limit for the numerator was needed.

Theorem 4.4 (Two root limits). *For every fixed real $a > 0$, $a^{1/n} \to 1$. Also $n^{1/n} \to 1$. Here $a^{1/n}$ is the positive power from standing fact E at exponent $1/n$; its nth power is a by the school power laws.*

Proof. First fix $a > 1$. Put $t_n = a^{1/n} - 1$. If this root were below 1, its positive integer power a would be below 1. Thus $t_n \ge 0$. Bernoulli gives

$$a = (1 + t_n)^n \ge 1 + nt_n, \qquad 0 \le t_n \le \frac{a - 1}{n}.$$

The upper bound tends to zero, so squeeze gives $t_n \to 0$ and hence $a^{1/n} \to 1$. For $a = 1$ the sequence is constant. If $0 < a < 1$, apply the proved case to $1/a > 1$. The reciprocal law, with denominator limit 1, then gives

$$a^{1/n} = \frac{1}{(1/a)^{1/n}} \longrightarrow 1.$$

These cases cover every $a > 0$.

For the variable-root limit, put $t_n = n^{1/(2n)} - 1$. A positive root below 1 would have its integer power below 1, contradicting $n \ge 1$. Thus $t_n \ge 0$. Bernoulli gives $(1 + t_n)^n \ge 1 + nt_n$. Both sides are nonnegative, so squaring gives

$$n = (1 + t_n)^{2n} \ge (1 + nt_n)^2 \ge n^2 t_n^2.$$

Thus $0 \le t_n \le 1/\sqrt{n}$. Given $\varepsilon > 0$, choosing $N > 1/\varepsilon^2$ makes $1/\sqrt{n} < \varepsilon$ for $n \ge N$. Squeeze gives $t_n \to 0$. The product and sum laws now yield

$$n^{1/n} = (1 + t_n)^2 \longrightarrow 1.$$

∎

4.2 Infinite limits

A real sequence may eventually exceed every prescribed real bound. We express this behaviour by an *infinite limit*. The notation $a_n \to +\infty$ means

$$\forall M \in \mathbb{R} \quad \exists N \in \mathbb{N} \quad \forall n \geq N : \quad a_n > M.$$

The notation $a_n \to -\infty$ means that for every $M \in \mathbb{R}$ all sufficiently late terms satisfy $a_n < M$. Neither symbol is a real number. The definition of a real limit in Lecture 3 still requires a real limit value.

For example, $n \to +\infty$: given $M \in \mathbb{R}$, the Archimedean property supplies $N > M$, and every $n \geq N$ exceeds M. Similarly, $\sqrt{n} \to +\infty$. Choose $N > \max\{M, 0\}^2$. Every $n \geq N$ then has $\sqrt{n} > \max\{M, 0\} \geq M$.

An unbounded sequence need not tend to either infinity. The sequence $(-1)^n n$ has arbitrarily large positive and negative terms. After every cutoff its odd terms are negative, so it cannot tend to $+\infty$. Its even terms prevent convergence to $-\infty$.

The limit laws for real limits cannot be applied with infinity as though it were a real number. In particular, the following patterns do not specify an answer. Each row shows two sequences with the same limiting pattern but different resulting limits.

Pattern	One outcome	Another outcome
$0/0$	$\frac{1/n}{1/n} = 1$	$\frac{1/n^2}{1/n} = 1/n \to 0$
∞/∞	$n/n = 1$	$n^2/n = n \to +\infty$
$\infty - \infty$	$(n+1) - n = 1$	$2n - n = n \to +\infty$
$0 \cdot \infty$	$(1/n)n = 1$	$(1/n^2)n = 1/n \to 0$

We call these *indeterminate forms*. The symbols describe the limits of components, rather than an arithmetic operation on infinity. Positive-base power patterns are examined with the standard limits in Lecture 6. The construction below will already give an example of 1^∞.

4.3 Monotone convergence

A real sequence is *nondecreasing* if $a_{n+1} \geq a_n$ for every $n \in \mathbb{N}$. It is *nonincreasing* if all these inequalities are reversed. A sequence with either property is called *monotone*. These definitions permit equality of consecutive terms.

Theorem 4.5 (Monotone convergence). *A nondecreasing real sequence that is bounded above converges to the supremum of its set of terms. A nonincreasing real sequence that is bounded below converges to the infimum of its set of terms.*

Proof. Suppose (a_n) is nondecreasing and bounded above. The set $A = \{a_n : n \in \mathbb{N}\}$ is nonempty and bounded above. By R, let $s = \sup A$. Given $\varepsilon > 0$, the number $s - \varepsilon$ is not an upper bound. Thus some a_N satisfies $s - \varepsilon < a_N \leq s$. For every $n \geq N$, monotonicity gives

$$s - \varepsilon < a_N \leq a_n \leq s.$$

Hence $|a_n - s| < \varepsilon$, proving $a_n \to s$.

If (a_n) is nonincreasing and bounded below, then $(-a_n)$ is nondecreasing and bounded above. The proved case supplies its limit. The scalar-multiple law gives a limit for (a_n). Negating the least upper bound of $\{-a_n\}$ gives the greatest lower bound of $\{a_n\}$, as asserted. ∎

The bound is essential. The sequence $a_n = n$ is nondecreasing and has no real limit. More generally, if a nondecreasing sequence is unbounded above, then it tends to $+\infty$. Given $M \in \mathbb{R}$, choose a term $a_N > M$. Every later term is at least a_N, so it also exceeds M.

A proof using monotone convergence has two separate obligations: establish the order of successive terms and supply a bound. Solving an equation for a possible limit does not establish either obligation. Sheet 4 applies this distinction to recursive sequences.

Example 4.6 (Proving existence before solving for a limit)**.** A sequence satisfies $a_1 = 0$ and $a_{n+1} = (2a_n + 5)/3$. We first prove that every term lies in $[0, 5]$. This holds at $n = 1$. If $0 \le a_n \le 5$, then

$$0 \le a_{n+1} = \frac{2a_n + 5}{3} \le \frac{10 + 5}{3} = 5.$$

Induction supplies the bound for all terms. It also gives

$$a_{n+1} - a_n = \frac{5 - a_n}{3} \ge 0.$$

Thus the sequence is nondecreasing and bounded above, so $a_n \to L$ for a real L. The shifted sequence a_{n+1} also tends to L: if $n \ge N$, then $n + 1 \ge N$, so the same error cutoff works. The limit laws applied to the recurrence now give

$$L = \frac{2L + 5}{3}, \qquad L = 5.$$

The fixed-point equation identifies the limit only after its existence has been proved.

4.4 Construction of the number e

Define two positive sequences by

$$u_n = \left(1 + \frac{1}{n}\right)^n, \qquad v_n = \left(1 + \frac{1}{n}\right)^{n+1}.$$

We will show that they converge to the same number from opposite sides. Only integer powers occur in these definitions.

First we prove that (u_n) increases. Rearranging its successive ratio and using Bernoulli with $h = -1/(n+1)^2$ gives

$$\frac{u_{n+1}}{u_n} = \left(1 - \frac{1}{(n+1)^2}\right)^n \frac{n+2}{n+1} \ge \left(1 - \frac{n}{(n+1)^2}\right) \frac{n+2}{n+1} = 1 + \frac{1}{(n+1)^3} > 1.$$

The value of h is greater than -1, so Bernoulli applies. The remaining factor is positive, so multiplication preserves the bound.

Next we prove that (v_n) decreases. This time Bernoulli with the positive number $h = 1/(n(n+2))$ gives

$$\frac{v_n}{v_{n+1}} = \left(1 + \frac{1}{n(n+2)}\right)^{n+1} \frac{n+1}{n+2} \ge \left(1 + \frac{n+1}{n(n+2)}\right) \frac{n+1}{n+2} = 1 + \frac{1}{n(n+2)^2} > 1.$$

Since the terms are positive, this proves $v_{n+1} < v_n$.

For each n, we also have $u_n < v_n$. Consequently, $u_n \le v_n \le v_1 = 4$. The increasing sequence (u_n) is therefore bounded above. Monotone convergence supplies its limit, denoted by e:

$$e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n. \tag{4.1}$$

This proves the existence of the number used in standing fact E. It does not prove the school-function properties assumed there.

The difference of the two sequences satisfies

$$0 < v_n - u_n = \frac{u_n}{n} \le \frac{4}{n}.$$

The squeeze theorem gives $v_n - u_n \to 0$, and the sum law now gives $v_n \to e$. For a fixed n, every $m \ge n$ satisfies $u_n \le u_m$ and $v_m \le v_n$. Letting $m \to \infty$ and applying Proposition 4.1 gives

$$u_n \le e \le v_n \qquad (n \in \mathbb{N}).$$

These are explicit lower and upper bounds for the same number. Since $u_2 = 9/4 > 2$ and $v_5 = 46656/15625 < 3$, we obtain $2 < e < 3$.

Both the bases of (u_n) and the constant bases 1 tend to 1, while their exponents n tend to $+\infty$. The former powers tend to $e > 2$ and the latter powers equal 1. Thus the power form 1^∞ is also indeterminate.

4.5 Subsequences and Bolzano–Weierstrass

A *subsequence* of (a_n) is a sequence (a_{n_k}) whose indices $n_1 < n_2 < \cdots$ are strictly increasing natural numbers. Induction gives $n_k \ge k$ for every $k \in \mathbb{N}$. If $a_n \to a$, then every subsequence converges to a: an original cutoff N works whenever $k \ge N$, since then $n_k \ge k \ge N$.

For example, the even-index subsequence of $(-1)^n$ is constantly 1. The odd-index subsequence is constantly -1. Both subsequences converge, although the original sequence does not. Having one convergent subsequence does not force the full sequence to converge.

Theorem 4.7 (Bolzano–Weierstrass). *Every bounded real sequence has a convergent subsequence. If all its terms lie in a closed interval $[a, b]$, the subsequence's limit also belongs to $[a, b]$.*

The proof repeatedly halves an interval and keeps a half containing infinitely many indices of the sequence. Figure 4.1 shows one possible chain of such intervals.

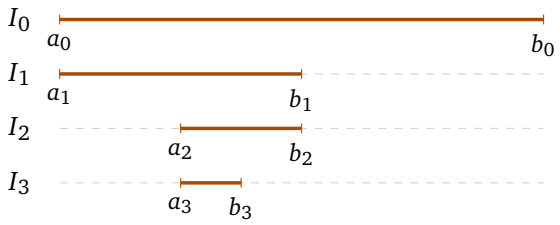


Figure 4.1: Nested closed halves. The left endpoints do not decrease, the right endpoints do not increase, and the lengths halve at each step.

Proof. Let (x_n) be the bounded sequence. Choose a closed interval $I_0 = [a_0, b_0]$ of positive length containing all terms. At least one of its closed halves contains terms at infinitely many indices. Otherwise both halves would contain terms at only finitely many indices, and so would their union. Keep a half with infinitely many indices and call it I_1.

Repeat this choice in each retained interval. We obtain nested intervals $I_k = [a_k, b_k]$ with infinitely many sequence indices in each and

$$b_k - a_k = \frac{b_0 - a_0}{2^k}.$$

The left endpoints form a nondecreasing sequence bounded above by b_0. By monotone convergence, $a_k \to c$ for some $c \in \mathbb{R}$. The displayed lengths tend to zero by the geometric limit of Lecture 3. Thus $b_k \to c$ by the sum law.

Choose n_1 with $x_{n_1} \in I_1$. Having chosen n_{k-1}, choose $n_k > n_{k-1}$ with $x_{n_k} \in I_k$. This is possible because I_k contains terms at infinitely many indices. The squeeze theorem applied to

$$a_k \le x_{n_k} \le b_k$$

gives $x_{n_k} \to c$. This is the required convergent subsequence. Finally, if its terms lie in $[a, b]$, order passes to limits and gives $a \le c \le b$. ∎

The endpoints a_k above are auxiliary bounds, while x_{n_k} denotes selected terms of the original sequence. Repetition of a sequence value is allowed. The construction requires infinitely many indices, rather than infinitely many different values.

The root limits above will be used in series tests. Sheet 4 studies recursive sequences and gives the Cauchy criterion with its complete proof.

Summary 4.

- Proposition 4.1 and Theorem 4.2: eventual order passes to limits; a sequence squeezed between two sequences with the same real limit converges to that limit.
- Theorem 4.4: $a^{1/n} \to 1$ for fixed $a > 0$, and $n^{1/n} \to 1$.
- Theorem 4.5: nondecreasing bounded-above sequences converge to their supremum. Nonincreasing bounded-below sequences converge to their infimum. Completeness R supplies the bound's limiting value.
- Equation (4.1): $u_n = (1+1/n)^n$ increases, $v_n = (1+1/n)^{n+1}$ decreases, and $u_n \le e \le v_n$, with $2 < e < 3$.
- Theorem 4.7: every bounded real sequence has a convergent subsequence. Limits of subsequences of a convergent sequence equal its limit. Closed-interval bounds survive the limit.

Lecture 5 uses these results to connect function limits with sequence limits. The subsequence argument will also control continuous functions on closed intervals in Lecture 25.

References

R. G. Bartle and D. R. Sherbert, *Introduction to Real Analysis*, 4th ed., Wiley, 2011, §§3.2–3.5.
S. R. Ghorpade and B. V. Limaye, *A Course in Calculus and Real Analysis*, 2nd ed., Springer, 2018, Chapter 2.